

Ahmed Farag

Data Scientist | Machine Learning Engineer | AI Engineer

Alexandria, Egypt | af0112651@gmail.com | +201120577817 | +201126516608

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[LinkedIn](#)

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with hands-on experience in data analysis, machine learning, predictive modeling, recommendation systems, and AI applications. Proficient in Python, SQL, Pandas, NumPy, and Scikit-learn, with experience in feature engineering, model evaluation, and ML deployment using FastAPI. Technical focus includes Machine Learning, Deep Learning, Neural Networks, NLP, Transformers, and Large Language Models (LLMs), with experience developing end-to-end data and AI solutions.

TECHNICAL SKILLS

Programming: Python, SQL, Java, C

Data Science: Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing, Statistical Analysis, Data Visualization, Feature Engineering, Feature Selection, Data Transformation

Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Ensemble Learning, Model Selection, Cross-Validation, Hyperparameter Tuning, Model Evaluation

Deep Learning & AI: Neural Networks, Deep Learning, CNNs, Transfer Learning, TensorFlow, PyTorch, Keras

NLP: Text Preprocessing, TF-IDF, Text Classification, Word Embeddings, Transformers, LLM Fundamentals, RAG

Databases: MySQL, MongoDB, PostgreSQL, SQLAlchemy

Backend: FastAPI, REST APIs, JWT Authentication, Redis, Docker

Big Data: Apache Hadoop, HDFS, Apache Kafka, Apache Spark

Tools: Git, GitHub, VS Code, Jupyter Notebook

EDUCATION

Alexandria University

Bachelor of Computer and Data Science | GPA: 3.43/4.00

Sep 2022 – Jul 2026

PROJECTS

1. Property Hub AI (Graduation Project)

AI-Powered Real Estate Platform — PropertyHUB

- Developed a full-stack AI-powered real estate platform using React, FastAPI, MySQL, and Redis.
- Built a hybrid recommendation system combining Content-Based Filtering and Learning-to-Rank techniques, trained using user interaction data from 200+ users.
- Developed a property price prediction model using 26,000+ real estate listings, achieving an R^2 score of 0.86.
- Designed and integrated RESTful APIs for authentication, property search, personalized recommendations, analytics, and price prediction. 

2. Customer Churn Prediction

Machine Learning Prediction System

- Developed a machine learning solution to predict customer churn using Python and Scikit-learn on the Telco Customer Churn dataset.
- Performed data preprocessing, feature engineering, and class balancing using SMOTE on 7,043 customer records.
- Trained and evaluated five classification models, with Random Forest achieving the best performance at 80.3% accuracy.
- Evaluated model performance using Accuracy, Precision, Recall, and F1-score to support customer retention analysis. 

PROFESSIONAL DEVELOPMENT

- Machine Learning Training — National Telecommunication Institute (NTI)
- AI & Machine Learning Summer Training — MAIM Company
- Fundamentals of Deep Learning — NVIDIA
- Introduction to Data Science — CISCO
- Computer Networks Training — Abu Qir Electricity Distribution Company
- IoT Summer Training — Faculty of Computers and Data Science

Languages

Arabic — Native | English — Professional Working Proficiency